



CMN 224 – SOFTWARE ENGINEERING
SRS DOCUMENT
ESTIMATION AND WORKSCHEDULE
CLIENT: WAGHI VALLEY COOPERATION

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GROUP 1

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Part 1: SRS Document

1.1 Purpose & Scope

1.1.1 Purpose

The purpose of the proposed system is to replace the cooperative's carbon-copy book records used at the roadside depot with a controlled digital record of member deliveries and payments.

The system shall provide a reliable record of transactions so that the cooperative can reduce disputes caused by missing or incomplete paper records and maintain a clear history of what each member delivered and what the cooperative owes or has paid each member.

1.1.2 In Scope

The system shall cover:

- Member identification and member records.
- Recording coffee and vegetable purchases at the depot.
- Recording the quantity and relevant transaction details for each purchase.
- Recording the member's payment amount.
- Recording fortnightly payment information.
- Viewing a member's transaction and payment history.
- Identifying transactions that have not yet been paid.
- Recording payment status.
- Correcting transaction records under the required approval rules.
- Recording disputes relating to transactions or payments.
- Providing records for the depot clerk, manager and treasurer.

1.1.3 Out of Scope

The following are outside the scope of this system:

- Managing the cooperative's farming activities.
- Managing crop production.
- Managing transport of coffee or vegetables.
- Managing staff payroll.
- Managing bank accounts or banking transactions.
- Managing loans or member financing.
- Managing inventory outside the coffee and vegetable purchases recorded at the depot.
- Providing internet or mobile network infrastructure.
- Providing mains electricity to the depot.
- Replacing the cooperative's existing payment policy.
- Making payment-related policy decisions on behalf of the member commit

1.2 Business Context and Stakeholders

The cooperative has 300 members and purchases coffee and vegetables at a roadside depot. Transactions are currently recorded in hard-copy books (notebooks). Members are paid fortnightly, and disputes are common because paper records can go missing.

The depot has no main power supply, with power supplied through solar and generator. The cooperation has no fixed internet connection and mobile coverage is patchy. The cooperative's town office is approximately 20 minutes away and has power and 4G connectivity.

1.2.1 Stakeholders

Stakeholder	Role / Interest	What the stakeholder needs
Members	Supply coffee and vegetables and receive payment	Accurate records of their deliveries, amounts owed, payments made and transaction history
Depot Clerk	Main user who records depot transactions	A straightforward way to record purchases accurately and retrieve previous records
Manager	Oversees cooperative operations	Visibility of depot records, transaction history, disputes and payment information
Treasurer	Responsible for payment-related records	Accurate amounts owed, payment status, fortnightly payment information and records supporting payment processing
Member Committee	Approves changes that affect payments	Sufficient information to review and approve payment-affecting changes
Cooperative	Organisation responsible for purchasing and paying members	Reliable records that reduce missing-record disputes and support accurate payment administration

Table 1

1.2.2 Primary User

The **depot clerk** is the primary system user because the clerk is responsible for recording purchases at the roadside depot.

1.2.3 Business Problem

The current carbon-copy recording process creates a risk that transaction records will be lost or become unavailable. When records are missing, members and cooperative staff can disagree about what was delivered or paid.

The system therefore needs to establish a consistent record of depot transactions and payments that can be retrieved when required.

1.3 Functional Requirements

The following are requirements that the system must maintain to satisfy what the client needs and how the system should implement the clients needs.

1.3.1 Member and Transaction Records

FR-001 – The system shall allow the depot clerk to select an existing cooperative member before saving a coffee or vegetable purchase transaction.

FR-002 – The system shall allow the depot clerk to record a purchase containing the member, produce type (coffee or vegetable), quantity, quality(grade), transaction date and amount payable.

FR-003 – The system shall assign a unique transaction identifier to every saved purchase.

FR-004 – The system shall allow the depot clerk, manager and treasurer to retrieve all saved transactions for a selected member.

1.3.2 Payments

FR-005 – The system shall assign each saved transaction one of two payment statuses: Unpaid or Paid.

FR-006 – The system shall allow the treasurer to record a member payment containing the member, payment date, payment amount and payment period.

FR-007 – The system shall allow the treasurer to associate a recorded payment with one or more unpaid transactions belonging to the same member.

FR-008 – The system shall calculate and display the total amount recorded as unpaid for a selected member from that member's unpaid transactions.

1.3.3 Disputes and Corrections

FR-009 – The system shall allow an authorised user to create a dispute containing the member, related transaction or payment, dispute date and description of the dispute.

FR-010 – The system shall allow an authorised user to submit a correction to an existing transaction and shall record the original transaction value, proposed new value and reason for the correction.

FR-011 – The system shall mark a correction as Pending Committee Approval when the proposed correction changes the amount payable or payment status of a member.

FR-012 – The system shall allow an authorised user to record either Approved or Rejected for a payment-affecting correction and shall record the decision date.

1.3.4 User Access and Records

FR-013 – The system shall provide separate access roles for depot clerk, manager and treasurer, with each role limited to the functions assigned to that role.

FR-014 – The system shall save a transaction entered at the depot when no network connection is available and shall retain that transaction until it is successfully transferred to the central record.

FR-015 – The system should transfer unsynchronised depot transactions to the central record when a network connection becomes available.

FR-016 – The system shall prevent a payment-affecting correction with a status of Pending Committee Approval or Rejected from changing the member's approved payment record.

1.4 Non-Functional Requirements

The following requirements are limited to the stated delivery conditions of the cooperative.

NFR-001 – The system shall operate in the depot environment where main electricity is unavailable, and power is supplied by solar power and a generator.

NFR-002 – The system shall support depot operation where fixed internet is unavailable, and mobile connectivity may be patchy.

NFR-003 – The system shall allow required depot transaction recording to continue when the depot has no usable network connection.

NFR-004 – The system shall support use of the system at the town office where power and 4G connectivity are available.

NFR-005 – Information recorded at the depot while connectivity is unavailable shall remain available for subsequent transfer or synchronisation when connectivity becomes available.

1.5 Constraints & Business Rules

1.5.1 Constraints

C-001 – The system shall be designed for the cooperative's current membership of **300 members**.

C-002 – The primary transaction-recording environment is the roadside depot.

C-003 – The depot has no mains power. Available power is solar power and generator power.

C-004 – The depot has no fixed internet connection and mobile signal is patchy.

C-005 – The town office is approximately **20 minutes from the depot** and has power and 4G connectivity.

C-006 – The depot clerk is the main user of the system.

1.5.2 Business Rules

BR-001 – Every coffee or vegetable purchase recorded by the cooperative shall be associated with the relevant cooperative member.

BR-002 – Members are paid on a **fortnightly** basis.

BR-003 – A recorded transaction shall remain associated with the member and payment information to which it relates.

BR-004 – A dispute concerning a transaction or payment shall be recorded against the relevant record where applicable.

BR-005 – Any change that affects a member's payment shall require approval by the **member committee**.

BR-006 – A payment-affecting change shall not be treated as an approved payment change until the required member committee approval has been recorded.

BR-007 – The depot clerk records depot transactions; the manager oversees cooperative records; and the treasurer handles payment-related records.

BR-008 – The system shall serve as the cooperative's digital record for transactions entered in the system, reducing reliance on the carbon-copy books for retrieval of those records.

1.6 Glossary

Term	Plain-language definition
Cooperative	The organisation that buys coffee and vegetables from its members and pays them.
Member	A person belonging to the cooperative who supplies coffee or vegetables.
Depot	The roadside location where the cooperative receives coffee and vegetables from members and records purchases.
Depot clerk	The staff member who is the main user of the system and records depot transactions.
Manager	The staff member responsible for overseeing cooperative operations and records.
Treasurer	The staff member responsible for payment-related records and information.
Member committee	The committee that must approve changes affecting member payments.
Transaction	A recorded purchase of coffee or vegetables from a cooperative member.
Payment	Money recorded as paid to a member for the member's purchases.
Fortnightly	Occurring once every two weeks; this is the cooperative's member payment cycle.
Payment status	The recorded state showing whether an amount has been paid or remains unpaid.
Dispute	A disagreement concerning a recorded transaction or payment.
Correction	A proposed or recorded change to an existing transaction or payment-related record because an error has been identified.
Payment-affecting change	A change that would alter the amount payable to a member or the payment status of a transaction.
Approval	The recorded acceptance by the member committee of a change that affects a member's payment.

Table 2

Part 2: Estimation and work schedule

2.1 Task Breakdown

#	Task	Description
1	Project/repository and offline database setup	Application structure, local database, backup/storage configuration
2	Member records	Create/search/update member records and member ID
3	Produce intake entry	Record member, crop/product, quantity, date and depot information
4	Pricing and payment calculation	Calculate value from produce quantity and applicable price
5	Payment recording and receipt	Record payment, assign transaction/receipt number and produce printable receipt
6	Intake/payment validation	Validation, duplicate prevention and basic error handling
7	Unit and integration tests	Test member, intake, calculation, payment and receipt functions
8	Power-failure/durability testing	Test interrupted writing, restart/recovery and data persistence
9	Code review, client review and rework	Review implementation with manager and correct issues
10	Documentation, installation and handover	User instructions, backup procedure, installation and staff handover

Table 3

2.2 Work Estimates

The figures in the table below are all estimates measured in hours.

$$E = (O + 4M + P) \div 6 \quad SD = (P - O) \div 6$$

#	Task	O	M	P	E	SD	Assumption behind it

1	Project/repository + offline database setup	3	5	10	5.50	1.17	Local database and offline-first structure work as planned
2	Member records	4	7	12	7.00	1.33	Around 300 members; simple CRUD/search
3	Produce intake entry	5	9	16	9.17	1.83	Coffee and vegetables require different fields/units
4	Pricing + payment calculation	4	7	14	7.67	1.67	Pricing rules remain relatively simple
5	Payment recording + receipt	5	9	16	9.17	1.83	Receipt numbering/printing must work without internet
6	Validation + duplicate prevention	3	5	10	5.50	1.17	Basic transaction validation only
7	Unit + integration tests	4	7	12	7.00	1.33	Covers core intake/payment workflows
8	Power-failure/durability testing	3	5	12	5.83	1.50	Generator/solar interruptions need realistic testing
9	Code/client review + rework	4	7	14	7.67	1.67	Manager feedback may require workflow changes
10	Documentation + installation + handover	3	6	12	6.50	1.50	Includes back-up instructions and staff training
Total					71.00	4.89h	
					h		

Table 4

2.3 Schedule

All three increments have visible milestones that are testable and the client can see. The weeks that are seen below is greatly influenced by the knowing fact that the team will have 10 hours in a week to work on the project. Using $E - SD = 65.7 \div 10 = 6.6$ weeks and $E + SD = 76.3 \div 10 = 7.6$ weeks. Estimating a total of 7 to 9 weeks.

Increment	What it delivers	Milestone the client can see	Weeks
1. Offline records and payments	Member records, produce intake, price calculation, payment records, receipt numbering, local storage and basic backup	Manager records a real member's coffee/vegetable delivery and payment, then finds the saved transaction and produces its receipt without internet.	1 – 7
2. Cooperative operations and reporting	Search/history, member transaction history, payment summaries, produce totals, reporting/export and improved backup/recovery	Manager selects a member and can show that member's recorded deliveries, payments and current balance for a chosen period.	8- 10
3. Deployment, refinement and handover	Final testing, deployment at the depot, staff training, documentation and operational handover	Depot staff complete a real intake-to-payment transaction on the deployed system and recover/view the record after restarting the equipment.	11-12

Table 5

2.4 Dependency Triangle Analysis

The project management triangle for this system consists of three key constraints: scope, cost and time, which must be balanced throughout the project. The **scope** defines what the system is expected to deliver, including member records, produce intake, pricing and payment calculations, payment recording, receipt generation, validation, testing, backup and recovery, documentation, installation and staff handover. The **cost** represents the resources required to complete the project and is estimated at approximately **71 development hours** based on the work estimates for all ten tasks. The **time** constraint is the planned **9-week development schedule**, covering the implementation of offline records and payments, cooperative operations and reporting, followed by deployment, refinement and handover. These three constraints are interdependent, meaning a change in one can affect the others. For example, increasing the project scope by adding new features may require additional development hours or extend the schedule. Similarly, reducing the available development time may require reducing the scope or allocating additional resources. Therefore, the project must maintain a practical balance between the required system features, available resources and the 12-week deadline to ensure the planned deliverables can be completed successfully.

Declarative use of AI

Tool: ChatGPT.

Purpose: Analysing final draft, and research to better understand realistic time frames on such existing work.